
Front matter

title: "Отчёта по лабораторной работе №1" subtitle: "Практическое научное письмо" author: "Конс Сирики"

Generic options

lang: ru-RU toc-title: "Содержание"

Bibliography

bibliography: bib/cite.bib csl: pandoc/csl/gost-r-7-0-5-2008-numeric.csl

Pdf output format

toc: true # Table of contents toc-depth: 2 lof: true # List of figures lot: true # List of tables fontsize: 12pt
linestretch: 1.5 papersize: a4 documentclass: scrreprt

l18n polyglossia

polyglossia-lang: name: russian options: - spelling=modern - babelshorthands=true polyglossia-otherlangs:
name: english

l18n babel

babel-lang: russian babel-otherlangs: english

Fonts

mainfont: IBM Plex Serif romanfont: IBM Plex Serif sansfont: IBM Plex Sans monofont: IBM Plex Mono
mathfont: STIX Two Math mainfontoptions: Ligatures=Common,Ligatures=TeX,Scale=0.94
romanfontoptions: Ligatures=Common,Ligatures=TeX,Scale=0.94 sansfontoptions:
Ligatures=Common,Ligatures=TeX,Scale=MatchLowercase,Scale=0.94 monofontoptions:
Scale=MatchLowercase,Scale=0.94,FakeStretch=0.9 mathfontoptions:

Biblatex

biblatex: true biblio-style: "gost-numeric" biblatexoptions:

- parenttracker=true
- backend=biber
- hyperref=auto
- language=auto
- autolang=other*
- citestyle=gost-numeric

Pandoc-crossref LaTeX customization

figureTitle: "Рис." tableTitle: "Таблица" listingTitle: "Листинг" lofTitle: "Список иллюстраций" lotTitle: "Список таблиц" lolTitle: "Листинги"

Misc options

indent: true header-includes:

- `\usepackage{indentfirst}`
- `\usepackage{float} # keep figures where there are in the text`
- `\floatplacement{figure}{H} # keep figures where there are in the text`

Цель работы

Целью данной лабораторной работы является установка TeXLive

Теоретическое введение

Installing TeXlive

TeX Live — the most complete LaTeX distribution supported by the TeX community. • Supports a large number of operating systems. • Developed since 1996. • Based on the teTeX distribution. • MacTeX — a variant for MacOS. • Main page: <https://www.tug.org/texlive/>. • TeX Live — is a distribution with continuous updates as part of the annual version of the distribution.

Installation from distribution packages [@lections]

- Ubuntu: `apt install texlive-full`
- Windows. Use the Chocolatey package manager. `choco install texlive`

Manual installation

• Links on the site are to mirrors. The mirror is selected automatically. • Download the installer: • Unix: `https://mirror.ctan.org/systems/texlive/tlnet/install-tl-unx.tar.gz cd /tmp/ wget https://mirror.ctan.org/systems/texlive/tlnet/install-tl-unx.tar.gz` • Windows: `https://mirror.ctan.org/systems/texlive/tlnet/install-tl-windows.exe` • For Windows: run the executable file and install. • For Linux • Unpack the downloaded file: `tar xzvf install-tl-unx.tar.gz` • Go to the unpacked directory and run the installer: `cd install-tl-[0-9]* sudo ./install-tl` • It is recommended to create links to executable files in the `/usr/local/bin` directory. To do this, in the console version of the utility, select the O option, and then L. To return to the previous menu, use R.

Updating to the next version of TeXlive

It is recommended to install the new version of TeX Live separately.

• But you can do a manual update using an existing installation. • Let's assume that our architecture is `x86_64-linux`. • If you have installed symbolic links to system directories (via installer option or `tlmgr path`

add), remove them:

1.1 Installing TeX Live

```
tlmgr path remove
```

- Move the entire TeX Live directory to match the new version, for example:

```
mv /usr/local/texlive/2024/ /usr/local/texlive/2025
```

- Remove package backups:

```
rm /usr/local/texlive/2025/tlpkg/backups/*
```

- Create links to executables:

```
/usr/local/texlive/2025/bin/x86_64-linux/tlmgr path add
```

- Download the latest version of the script update-tlmgr-latest.sh:

```
wget https://mirror.ctan.org/systems/texlive/tlnet/update-tlmgr-latest.sh  
\  
-O /tmp/update-tlmgr-latest.sh
```

- Run the script:

```
sh /tmp/update-tlmgr-latest.sh --upgrade
```

- If you do not want to use the default repository for downloading new files, then replace it:

```
tlmgr option repository <reponame>
```

- Update the TeX Live package manager:

```
tlmgr update --self
```

- Update TeX Live packages:

```
tlmgr update --all
```

- Set symbolic links to executables in system directories (/usr/local/bin):

```
tlmgr path add
```

1 LaTeX basics

- You can recreate the cache luatex under the user:

```
mv ~/.texlive2024 ~/.texlive2025
luaotfload-tool -fu
```

- If you don't do this, the cache will be recreated on the first run of luatex.

Выполнение лабораторной работы

Проверка установка teXlive

(см. Рис. [-@fig:001]).

```
[nadiaezzakate@Nadias-MacBook-Air ~ % pdflatex --version
pdfTeX 3.141592653-2.6-1.40.27 (TeX Live 2025)
kpathsea version 6.4.1
Copyright 2025 Han The Thanh (pdfTeX) et al.
There is NO warranty. Redistribution of this software is
covered by the terms of both the pdfTeX copyright and
the Lesser GNU General Public License.
For more information about these matters, see the file
named COPYING and the pdfTeX source.
Primary author of pdfTeX: Han The Thanh (pdfTeX) et al.
Compiled with libpng 1.6.46; using libpng 1.6.46
Compiled with zlib 1.3.1; using zlib 1.3.1
Compiled with xpdf version 4.04
```

{ #fig:001 width=100% }

```
-----  
[nadiaezzakate@Nadiaz-MacBook-Air ~ % tex --version  
TeX 3.141592653 (TeX Live 2025)  
kpathsea version 6.4.1  
Copyright 2025 D.E. Knuth.  
There is NO warranty.  Redistribution of this software is  
covered by the terms of both the TeX copyright and  
the Lesser GNU General Public License.  
For more information about these matters, see the file  
named COPYING and the TeX source.  
Primary author of TeX: D.E. Knuth.  
[nadiaezzakate@Nadiaz-MacBook-Air ~ % tlmgr --version  
tlmgr revision 74241 (2025-02-24 00:10:34 +0100)  
tlmgr using installation: /usr/local/texlive/2025  
TeX Live (https://tug.org/texlive) version 2025
```

{ #fig:001 width=100% }

Выводы

Таким образом, была достигнута цель установил TeXlive.

Список литературы{.unnumbered}
